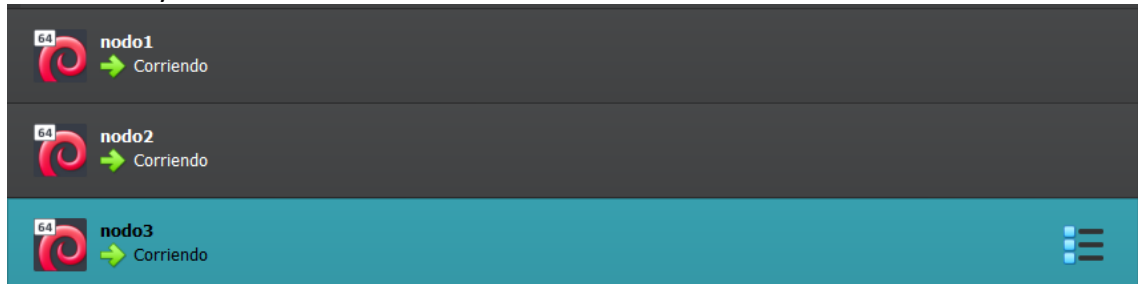


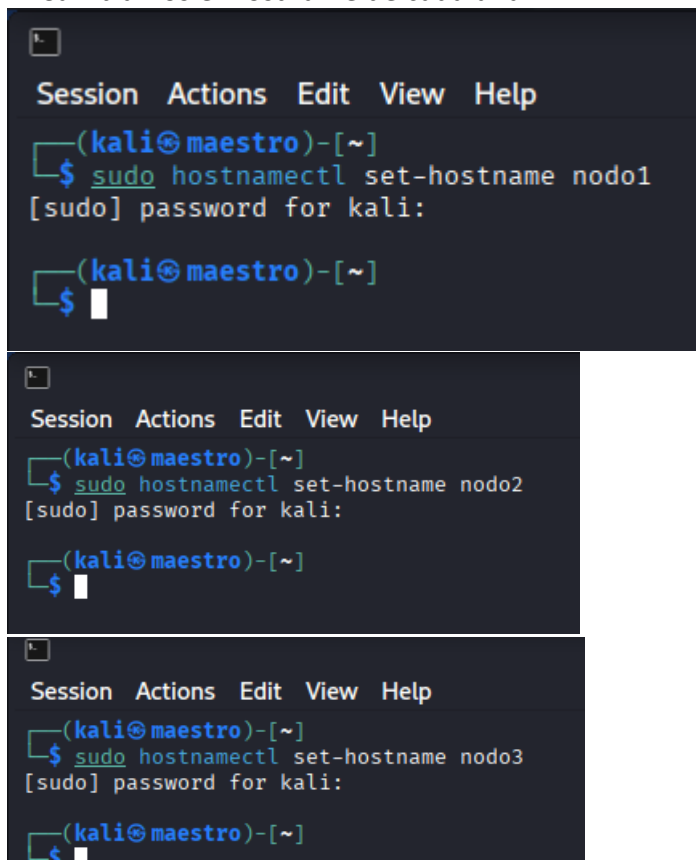
Solucion tarea 7

0. Creamos las 3 máquinas necesarias.

- Nodo 1: Es la máquina maestra.
- Nodo 2 y Nodo 3: Serán los dos esclavos



1. Cambiamos el hostname de cada una



2. Editamos el archivo `/etc/hosts` para que las máquinas se reconozcan entre sí.
- Lo abrimos con el comando `sudo nano /etc/hosts`

```
Session  Actions  Edit  View  Help
GNU nano 8.6
127.0.0.1    localhost

10.0.2.15  nodo1
10.0.2.15  nodo2
10.0.2.15  nodo3

::1         localhost ip6-localhost ip6-loopback
ff02::1     ip6-allnodes
ff02::2     ip6-allrouters
```

3. Generamos la clave ssh para que el nodo 1 se pueda conectar a los dos esclavos

```
(kali@nodo1)-[~]
$ ssh nodo2
Linux nodo1 6.12.38+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.38-1kali1 (2025-08-12) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Fri Jan 16 05:57:32 2026 from 10.0.2.15
(kali@nodo1)-[~]
$
```

4. Configuramos Hadoop

- core-site.xml: Editamos el archivo

```
GNU nano 8.6 core-site.xml
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->
<!-- Put site-specific property overrides in this file. -->

<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://nodo1:9000</value>
  </property>
</configuration>
```

- yarn-site.xml

```
GNU nano 8.6
<?xml version="1.0"?>
<!--
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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->

<configuration>
  <property>
    <name>yarn.resourcemanager.hostname</name>
    <value>nodo1</value>
  </property>

  <property>
    <name>yarn.nodemanager.aux-services</name>
    <value>mapreduce_shuffle</value>
  </property>
</configuration>
```

- hdfs-site.xml

```
Session Actions Edit View Help
GNU nano 8.6 hdfs-site.xml
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
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limitations under the License. See accompanying LICENSE file.
-->
<!-- Put site-specific property overrides in this file. -->

<configuration>
  <!-- directorios locales para NameNode y DataNode -->
  <property>
    <name>dfs.namenode.name.dir</name>
    <value>file:///datos/hadoop/namenode</value>
  </property>

  <property>
    <name>dfs.datanode.data.dir</name>
    <value>file:///datos/hadoop/datanode</value>
  </property>

  <!-- en pseudocluster: replicación 1 -->
  <property>
    <name>dfs.replication</name>
    <value>2</value>
  </property>
</configuration>
```

- workers

```
GNU nano 8.6 workers *
nodo2
nodo3
```

5. Copiamos la configuración del paso 4 en los nodos 2 y 3

- Lo hacemos para el nodo 2

```
(kali@nodo1)~[~]
$ scp -r /home/kali/Desktop/hadoop/etc/hadoop/* nodo2:/home/kali/Desktop/hadoop/etc/hadoop/
capacity-scheduler.xml
configuration.xml
container-executor.cfg
core-site.xml
hadoop-env.cmd
hadoop-env.sh
hadoop-metrics2.properties
hadoop-policy.xml
hadoop-user-functions.sh.example
hdfs-rbf-site.xml
hdfs-site.xml
https-env.sh
https-log4j.properties
https-site.xml
kms-acls.xml
kms-env.sh
kms-log4j.properties
kms-site.xml
log4j.properties
mapred-env.cmd
mapred-env.sh
mapred-queues.xml.template
mapred-site.xml
example.sh
ssl-client.xml.example
ssl-server.xml.example
user_ec_policies.xml.template
workers
yarn-env.cmd
yarn-env.sh
yarnservice-log4j.properties
yarn-site.xml
```

Lo hacemos para el nodo 3

```
(kali@nodo1)~[~]
$ scp -r /home/kali/Desktop/hadoop/etc/hadoop/* nodo3:/home/kali/Desktop/hadoop/etc/hadoop/
capacity-scheduler.xml
configuration.xml
container-executor.cfg
core-site.xml
hadoop-env.cmd
hadoop-env.sh
hadoop-metrics2.properties
hadoop-policy.xml
hadoop-user-functions.sh.example
hdfs-rbf-site.xml
hdfs-site.xml
https-env.sh
https-log4j.properties
https-site.xml
kms-acls.xml
kms-env.sh
kms-log4j.properties
kms-site.xml
log4j.properties
mapred-env.cmd
mapred-env.sh
mapred-queues.xml.template
mapred-site.xml
example.sh
ssl-client.xml.example
ssl-server.xml.example
user_ec_policies.xml.template
workers
yarn-env.cmd
yarn-env.sh
yarnservice-log4j.properties
yarn-site.xml
```

```
(kali㉿nodo1)-[~/Desktop/hadoop/sbin]
$ /home/kali/Desktop/hadoop/sbin/start-yarn.sh
Starting resourcemanager
Starting nodemanagers
```

Comprobamos que se han iniciado todos los servicios correctamente

- En el nodo1

```
(kali@nodo1)-[~]
$ jps
8624 DataNode
9641 Jps
9311 NodeManager
```

- En el nodo 2 y 3 aparece

```
(kali@nodo1)-[~]
$ jps
9472 NodeManager
10077 Jps
9086 DataNode
```

9. Utilizamos MapReduce

- Creamos el directorio en HDFS

```
(kali@nodo1)-[~]
$ /home/kali/Desktop/hadoop/bin/hdfs dfs -mkdir /prueba
```

- Creamos y subimos el archivo prueba.txt

```
(kali@nodo1)-[~]
$ echo "Yo ofrezco
grandes, vírgenes, intactas y sencillas,
para mis delicias y el placer de mis amigos,
estas muchas dadas variadas, soladas y traduccidas sobre su tierra natal y sobre el agua
ellas me fueron dulces durante los ocios en remotas mares, bajo un cielo ahora lejano.
Por eso las doy.

Sencillas, sonrientes y llenas de ingenuidad, como la musulmana Schehrazada, su madre suculenta que las dió a luz en el misterio; fermentando con emoción en los brazos de un príncipe sublime -lúbrico y feroz-, bajo la mirada enternecida
a de Alah, clemente y misericordioso.

Al venir al mundo fueron delicadamente medidas por las manos de la lustral Donizrada, su buena tía, que grabó sus nombres sobre hojas de oro coloreadas de húmedas pederías y las cuidó bajo el terciopelo de sus pupilas hasta la adolescencia dura, para espacillas después, voluptuosas y libres, sobre el mundo oriental, eternizado por su sacras.

Yo es las entrego tales como son, en su frescor de carne y de rosa. Sólo existe un método honrado y lógico de traducción: la «literalidad», una literalidad impersonal, apenas atenuada por un leve parpadec y una ligera sonrisa del traductor. Ella crea, sugestiva, la más grande potencia literaria. Ella produce el placer de la evocación. Ella es la garantía de la verdad. Ella es firme e inmutable, en su desnudez de piedra. Ella cautiva el aroma primitivo y lo cristaliza. Ella separa y desata... Ella fija.

La literalidad encadena el espíritu divagador y lo doma, al mismo tiempo que detiene la infernal facilidad de la pluma. Yo me felicito de que así sea; porque ¿dónde encontrar un traductor de genio simple, anónimo, libre de la necia manía de su renombre?" > archivo.txt
$ /home/kali/Desktop/hadoop/bin/hdfs dfs -put archivo.txt /prueba
```

- Ejecutamos el WordCount

```
(kali@nodo1)-[~]
$ /home/kali/Desktop/hadoop/bin/hadoop jar \
/home/kali/Desktop/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar \
wordcount /prueba /salida_final

2026-01-18 18:31:41,476 INFO client.DefaultHARMFailoverProxyProvider: Connecting to ResourceManager at nodo1/192.168.1.170:8032
2026-01-18 18:31:42,182 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/kali/.staging/job_1768778996056_0001
2026-01-18 18:31:42,792 INFO input.FileInputFormat: Total input files to process : 1
2026-01-18 18:31:43,040 INFO mapreduce.JobSubmitter: number of splits:1
2026-01-18 18:31:43,530 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1768778996056_0001
2026-01-18 18:31:43,530 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-01-18 18:31:43,889 INFO conf.Configuration: resource-types.xml not found
2026-01-18 18:31:43,890 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-01-18 18:31:44,672 INFO impl.YarnClientImpl: Submitted application application_1768778996056_0001
2026-01-18 18:31:44,882 INFO mapreduce.Job: The url to track the job: http://nodo1:8088/proxy/application_1768778996056_0001/
2026-01-18 18:31:44,890 INFO mapreduce.Job: Running job: job_1768778996056_0001
2026-01-18 18:31:58,850 INFO mapreduce.Job: Job job_1768778996056_0001 running in uber mode : false
2026-01-18 18:31:58,852 INFO mapreduce.Job: map 0% reduce 0%
2026-01-18 18:32:07,288 INFO mapreduce.Job: map 100% reduce 0%
2026-01-18 18:32:16,543 INFO mapreduce.Job: map 100% reduce 100%
2026-01-18 18:32:17,578 INFO mapreduce.Job: Job job_1768778996056_0001 completed successfully
2026-01-18 18:32:17,732 INFO mapreduce.Job: Counters: 54
  File System Counters
    FILE: Number of bytes read=3136
    FILE: Number of bytes written=558849
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=2316
    HDFS: Number of bytes written=2145
    HDFS: Number of read operations=8
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Launched map tasks=1
    Launched reduce tasks=1
    Data-local map tasks=1
    Total time spent by all maps in occupied slots (ms)=5884
    Total time spent by all reduces in occupied slots (ms)=5921
    Total time spent by all map tasks (ms)=5884
    Total time spent by all reduce tasks (ms)=5921
    Total vcore-milliseconds taken by all map tasks=5884
    Total vcore-milliseconds taken by all reduce tasks=5921
    Total megabyte-milliseconds taken by all map tasks=6025216
    Total megabyte-milliseconds taken by all reduce tasks=6063104
```

Map-Reduce Framework

```
Map input records=7
Map output records=391
Map output bytes=3775
Map output materialized bytes=3136
Input split bytes=101
Combine input records=391
Combine output records=247
Reduce input groups=247
Reduce shuffle bytes=3136
Reduce input records=247
Reduce output records=247
Spilled Records=494
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=128
CPU time spent (ms)=2990
Physical memory (bytes) snapshot=586231808
Virtual memory (bytes) snapshot=5466726400
Total committed heap usage (bytes)=471859200
Peak Map Physical memory (bytes)=324190208
Peak Map Virtual memory (bytes)=2720837632
Peak Reduce Physical memory (bytes)=262041600
Peak Reduce Virtual memory (bytes)=2745888768
```

Shuffle Errors

```
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
```

File Input Format Counters

```
Bytes Read=2215
```

File Output Format Counters

```
Bytes Written=2145
```

10. Yarn avanzado

```
(kali@nodo1)-[~]
$ /home/kali/Desktop/hadoop/bin/yarn node -list
2026-01-18 18:34:08,907 INFO client.DefaultNoHARMAFailoverProxyProvider: Connecting to ResourceManager at nodo1/192.168.1.170:8032
Total Nodes:1
  Node-Id      Node-State Node-Http-Address  Number-of-Running-Containers
  nodo1:35121  RUNNING   nodo1:8042        0

(kali@nodo1)-[~]
$ /home/kali/Desktop/hadoop/bin/yarn application -list
2026-01-18 18:34:59,823 INFO client.DefaultNoHARMAFailoverProxyProvider: Connecting to ResourceManager at nodo1/192.168.1.170:8032
Total number of applications (application-types: [], states: [SUBMITTED, ACCEPTED, RUNNING] and tags: []) is 0
  Application-Id  Application-Name  Application-Type  User  Queue  State  Final-State  Progress  Tracking-URL
```

```

(kali@nod01)-[~]
$ /home/kali/Desktop/hadoop/bin/hadoop jar \
/home/kali/Desktop/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar \
wordcount \
-Dmapreduce.job.queueName=desa \
/prueba /salida_desa

2026-01-18 18:41:49,872 INFO client.DefaultNoHARMFaloverProxyProvider: Connecting to ResourceManager at nod01/192.168.1.170:8032
2026-01-18 18:41:50,463 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/kali/.staging/job_1768778996056_0003
2026-01-18 18:41:50,938 INFO input.FileInputFormat: Total input files to process : 1
2026-01-18 18:41:51,158 INFO mapreduce.JobSubmitter: number of splits:1
2026-01-18 18:41:51,544 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1768778996056_0003
2026-01-18 18:41:51,544 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-01-18 18:41:51,843 INFO conf.Configuration: resource-types.xml not found
2026-01-18 18:41:51,844 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-01-18 18:41:52,054 INFO impl.YarnClientImpl: Submitted application application_1768778996056_0003
2026-01-18 18:41:52,092 INFO mapreduce.Job: The url to track the job: http://nod01:8088/proxy/application_1768778996056_0003/
2026-01-18 18:41:52,095 INFO mapreduce.Job: Running job: job_1768778996056_0003
2026-01-18 18:42:00,273 INFO mapreduce.Job: Job job_1768778996056_0003 running in uber mode : false
2026-01-18 18:42:00,275 INFO mapreduce.Job: map 0% reduce 0%
2026-01-18 18:42:07,654 INFO mapreduce.Job: map 100% reduce 0%
2026-01-18 18:42:14,857 INFO mapreduce.Job: map 100% reduce 100%
2026-01-18 18:42:14,875 INFO mapreduce.Job: Job job_1768778996056_0003 completed successfully
2026-01-18 18:42:15,031 INFO mapreduce.Job: Counters: 54
  File System Counters
    FILE: Number of bytes read=3136
    FILE: Number of bytes written=558833
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=2316
    HDFS: Number of bytes written=2145
    HDFS: Number of read operations=8
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Launched map tasks=1
    Launched reduce tasks=1
    Data-local map tasks=1
    Total time spent by all maps in occupied slots (ms)=5153
    Total time spent by all reduces in occupied slots (ms)=4175
    Total time spent by all map tasks (ms)=5153
    Total time spent by all reduce tasks (ms)=4175
    Total vcore-milliseconds taken by all map tasks=5153
    Total vcore-milliseconds taken by all reduce tasks=4175
    Total megabyte-milliseconds taken by all map tasks=5276672
    Total megabyte-milliseconds taken by all reduce tasks=4275200

```

```

Total megabyte-milliseconds taken by all reduce
Map-Reduce Framework
  Map input records=7
  Map output records=391
  Map output bytes=3775
  Map output materialized bytes=3136
  Input split bytes=101
  Combine input records=391
  Combine output records=247
  Reduce input groups=247
  Reduce shuffle bytes=3136
  Reduce input records=247
  Reduce output records=247
  Spilled Records=494
  Shuffled Maps =1
  Failed Shuffles=0
  Merged Map outputs=1
  GC time elapsed (ms)=108
  CPU time spent (ms)=2390
  Physical memory (bytes) snapshot=536371200
  Virtual memory (bytes) snapshot=5451923456
  Total committed heap usage (bytes)=394264576
  Peak Map Physical memory (bytes)=310259712
  Peak Map Virtual memory (bytes)=2722406400
  Peak Reduce Physical memory (bytes)=226111488
  Peak Reduce Virtual memory (bytes)=2729517056
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=2215
File Output Format Counters
  Bytes Written=2145
```